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Optimizing carbon capture and storage (CCS) infrastructure development using a python tool for source-sink matching and cluster formation

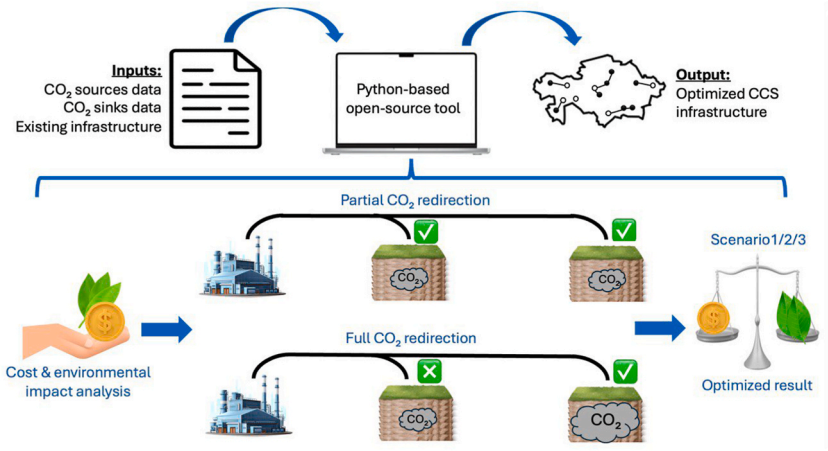
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HIGHLIGHTS

- First optimized CCS hubs in Central Asia using MILP modeling.
- Developed a new method with full and partial redirection for source-sink matching.
- Identified 17 storage sites with 5784 Mt CO_2 capacity.
- Found 100 km clustering minimizes cost, while 240–300 km balances emissions.
- Created an open-source Python tool for flexible CCS planning.

GRAPHICAL ABSTRACT



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ABSTRACT

The large-scale deployment of Carbon Capture and Storage (CCS) infrastructure is critical for Kazakhstan's decarbonization strategy, particularly in hard-to-abate industries. This study develops an optimization-driven framework for CCS infrastructure planning, focusing on source-sink matching and cost-effective clustering of emitters. A Python-based mixed-integer linear programming (MILP) model is applied to identify optimal CCS hub configurations by minimizing the total cost of capture, transportation, and storage while accounting for environmental impact. The results demonstrate that a 100 km clustering radius achieves the lowest cost scenario, with full redirection costing approximately $23.30 billion and partial redirection $24.02 billion. However, the CO_2 net abated is highest in this scenario, with emissions reaching 140.55 $MtCO_2$ and 142.73 $MtCO_2$, respectively. A 240–300 km clustering radius offers a more balanced trade-off, stabilizing costs at approximately $34.57 billion while reducing emissions. The analysis also reveals that capture costs represent the greatest portion of total CCS expenses (50.47 % at 100 km), while transportation costs increase with clustering radius, rising from 45.61 % to 61.01 % at 180 km. Leveraging existing pipeline infrastructure significantly improves cost

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efficiency, particularly in shorter clustering radii. The findings highlight the importance of integrating cost-efficient CCS hub networks and policy interventions, including regulatory frameworks, financial incentives, and international cooperation, to accelerate Kazakhstan's transition to a low-carbon economy. Future research should refine dynamic clustering models and incorporate real-time storage capacity assessments for improved decision-making. To enhance model repeatability, we provide an open-source Python code for future use.

1. Introduction

Carbon capture and storage (CCS) is a critical pathway for reducing anthropogenic CO_2 emissions, particularly in hard-to-abate sectors like cement, steel, and chemicals, which contribute over 20 % of global emissions (IPCC et al., 2023). As nations work toward the Paris Agreement's 1.5 °C target, the IPCC underscores CCS as essential for net-zero transitions, particularly in fossil fuel-dependent economies. However, global CCS deployment remains inadequate, capturing only 45 million tonnes of CO_2 annually—far short of the 7.6 gigatonnes needed by 2050 (Global CCS Institute, 2023). This gap highlights the urgency of scalable, cost-effective CCS infrastructure tailored to regional needs.

Kazakhstan exemplifies this challenge: coal supplies 80 % of its primary energy (Abuov et al., 2022), and industrial clusters drive high emissions. Its geography also presents potential for transboundary CCS collaboration, aligning with national climate commitments. Yet, past CCS infrastructure designs in Kazakhstan relied on manual, subjective approaches, limiting cost efficiency and scalability. Addressing these limitations requires systematic, optimization-driven tools to unlock CCS potential.

Transitioning from isolated CCS projects to shared "hub-and-cluster" systems offers significant economic and environmental benefits. Aggregating emissions from multiple sources—such as power plants, steel mills, and refineries—into centralized transport and storage networks reduces unit costs by up to 40 % compared to point-to-point systems (Middleton et al., 2020). For Kazakhstan, where emissions are regionally dispersed, shared infrastructure alleviates financial burdens and maximizes storage use in depleted oil fields and saline aquifers (Khoyashov et al., 2024). European CCS hubs like Norway's Northern Lights and the UK's Hynet emphasize flexible design for phased expansion and policy adaptation (Zero Emission Platform, 2021). However, earlier manual approaches to source-sink matching in Kazakhstan produced suboptimal network designs, overlooking pipeline routing costs, injection risks, and emissions. Robust computational tools are essential for balancing cost minimization, emissions reduction, and scalability while aligning with regional governance frameworks.

Optimizing CCS network design has evolved significantly since the early 2000s. The systematic pairing of CO_2 sources with geologic sinks emerged as a critical research frontier in the early 2000s, driven by the need to operationalize CCS at scale. One of the earliest conceptual frameworks was proposed by Benson and Ogden (2003), who developed a linear optimization model to evaluate the cost-effectiveness of transporting CO_2 from power plants to storage sites in Australia. Their work laid the groundwork for subsequent studies by formalizing the economic trade-offs among capture costs, pipeline distances, and storage site capacities. However, these early models operated under static assumptions—fixed emission rates, uniform storage lifetimes, and deterministic geologic conditions—limiting their applicability to real-world scenarios marked by variability and uncertainty. More recent advances have incorporated spatial and temporal complexities. Lee et al. (2014) introduced a unified model for retrofitting power plants with CCS, while Tan et al. (2012, 2013) used mixed-integer nonlinear programming (MINLP) to enable flexible scheduling of CO_2 injections. Meanwhile, Zhang et al. (2019) integrated uncertainty modeling, accounting for risks like fluctuating CO_2 prices and geologic storage failures. Building on these efforts, Olson and Yaw (2025), propose a novel MILP-based workflow that explicitly integrates probabilistic modeling of storage uncertainty to design robust CCS networks from the ground up, rather than merely adapting to or repairing underperforming infrastructure.

Parallel research has focused on optimizing CCS network architectures. Santibanez-Gonzalez (2017) examined CO_2 pricing mechanisms, while Sun et al., 2024; Sun and Chen, 2017 developed a pipeline network model for China's coal-heavy industry, emphasizing economies of scale. Ravi et al. (2017) applied mixed-integer linear programming to design a nationwide CCS network in the Netherlands. GIS-based tools like SimCCS (Middleton et al., 2020) optimized pipeline routes but relied on static storage assumptions, limiting applicability in emerging economies. Other models, such as InfraCCS (Knoope et al., 2013), lacked transboundary considerations. Region-specific tools like ChinaCCS and Sequestrix™ were limited by geologic and policy constraints. Fan et al. (2023) emphasized the need for open-source platforms integrating leakage risks and social acceptance metrics into CCS optimization. More comprehensive advancements in CCS network optimization have increasingly focused on multi-objective approaches and the integration of diverse supply chain elements. For instance, Zhang et al. (2023) developed a multi-objective mixed-integer linear programming (MILP) framework, incorporating a superstructure method with intermediate nodes, to optimize CCUS supply chains in Northeast China by minimizing both total annual cost and environmental impact (measured by Global Warming Potential via Life Cycle Assessment). This work highlighted the inherent trade-offs between economic and environmental objectives and the efficiency benefits of hub-like transport configurations. Further expanding the scope, Nie et al. (2023) applied a source-sink matching MILP model combined with LCA to analyze CCUS supply chains in coastal regions (Guangdong Province, China), specifically integrating CO2 utilization pathways (e.g., methanol synthesis) and offshore storage. Their findings revealed that unit abatement costs can increase with higher reduction targets due to storage complexities, and emphasized the significant role of carbon trading policies in enhancing economic viability, while also addressing uncertainties through Monte Carlo simulations.

Despite these advancements, key challenges persist. Most tools remain proprietary, costly, or rigidly region-specific. Even sophisticated models often neglect multi-objective optimization, forcing trade-offs between cost and environmental priorities. Existing multi-objective studies often do not take into account factors such as changing CO2 emission profiles and gradual depletion of storage capacity over time. A static model may not capture optimal long-term investment strategies or adaptation pathways needed for a truly resilient and sustainable CCUS network. Fan et al. (2023) emphasized the need for open-source platforms integrating leakage risks and social acceptance metrics into CCS optimization.

To address these gaps, we present a Python-based, open-source optimization tool for CCS network planning. Our framework introduces three key innovations:

- Scenario-Based Decision-Making: Users can compare two strategies for managing source-sink mismatches: (a) Partial redirection, where excess CO_2 is stored in secondary sinks, or (b) Full redirection, prioritizing sinks with sufficient capacity. This flexibility is critical for regions with uneven storage availability, like Kazakhstan. Considering CO_2 storage capacity and ease in adding new CO_2 (Tapia, 2024) sources allow users to model multi-period planning horizons for adaptive decision-making regarding infrastructure expansion, technology upgrades, and operational adjustments over several decades.

- Multi-Objective Optimization: Our model minimizes both transport costs and emissions using a weighted goal programming approach, unlike legacy tools that treat environmental impact as a secondary constraint.
- Open-Source Accessibility: Designed for ease of use, the tool requires minimal input data to generate optimized CCS networks. Its modular architecture ensures adaptability to different regions, eliminating reliance on proprietary software or region-specific calibrations, simultaneously allowing dynamic analysis. By prioritizing simplicity, it empowers stakeholders in developing economies to plan cost-effective CCS infrastructure.

Our approach builds on global case studies, including Norway's Northern Lights and Malaysia's offshore CCS clusters (Hatta et al., 2022), while addressing their limitations. We integrate Garnett et al.'s (2022) probabilistic storage capacity models into a user-friendly framework that accounts for evolving emission profiles, such as Kazakhstan's planned coal phaseouts. By removing manual workflows and subjective assumptions, as seen in our earlier Kazakhstan work (Khoyashov et al., 2024), our tool advances scalable, science-driven CCS deployment that is both affordable and adaptable.

2. Source-sink matching scenarios

In CCS hub design, clustering radius optimization depends on the geographic distribution and capacity of emission sources and storage sinks. Exploring different scenarios provides insights into how various parameters impact configurations.

Two primary methods manage source-sink relationships: full redirection (Fig. 1a) and partial redirection (Fig. 1b). Full redirection transfers all emissions from a full storage site to the nearest available sink, while partial redirection only redirects selected sources, keeping others at the initial site. Unlike full redirection, partial redirection allows splitting emission hubs, offering greater flexibility in balancing storage use, particularly in regions with uneven source-sink distributions.

Clustering is essential for organizing emission sources into efficient CCS hubs, enabling shared capture and storage infrastructure. This approach reduces operational costs and enhances scalability. The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is widely used for forming clusters of varying shapes and sizes while detecting outliers (Zhang et al., 2023). Flexible clustering and adjustable radii are crucial for CCS implementation, as sources may not fit predefined categories. In this study, DBSCAN was applied to the spatial coordinates of CO_2 sources using two key parameters: *eps*, which defines the maximum geographic distance between neighboring sources in a cluster, and *min_samples*, which sets the minimum number of points required to form a valid cluster. We set *min_samples* = 2 to allow small groups of emitters to form viable hubs, while varying eps to assess the effect of clustering radius on hub formation. After clustering, each group of sources was connected using a minimum spanning tree to minimize intra-cluster pipeline distances. This combined approach strikes a balance between clustering flexibility and computational efficiency, directly informing the routing, cost, and emission modeling of the CCS network.

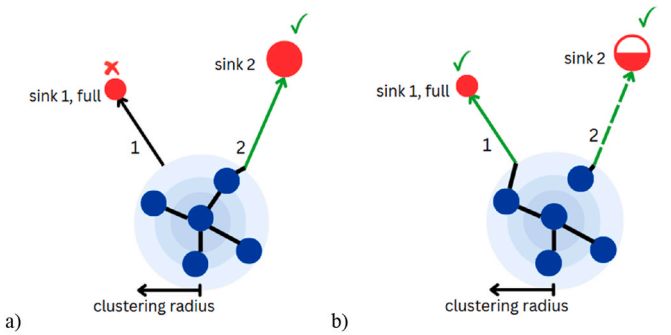


Fig. 1. The diagram of sink-to-source connection by full and partial redirections on clustered sources.

The effectiveness of clustering in CCS depends on emission densities and storage capacity. Northern Europe benefits from large clustering radii and full redirection systems, ensuring continuous sequestration during storage saturation (Northern Lights Project, 2020). In contrast, low-density regions like Australia (27 Mt CO_2 storage planned by 2030) use small clustering radii and localized storage, minimizing long-distance transport (Sivanathan, 2022). In regions with distant sinks, such as the U.S. Midwest, large clustering radii consolidate emissions for long-distance transport (Illinois Basin-Decatur Project), while partial redirection helps balance transport distances and infrastructure use (Smith et al., 2021). Offshore storage (e.g., North Sea) requires moderate to large clustering radii to optimize pipeline infrastructure. The trade-off is crucial: full redirection simplifies logistics but raises costs due to longer pipelines and compression, while partial redirection increases flexibility by distributing emissions across multiple sinks. Cost efficiency depends on pipeline layout, compression needs, and economies of scale. As CCS expands, adaptive redirection strategies will be key to balancing infrastructure investment and carbon storage efficiency.

Existing pipeline infrastructure plays a critical role in CCS hub development, significantly reducing capital expenditures and deployment time. Repurposing pipelines lowers infrastructure costs, avoids regulatory delays, and enables faster project implementation. In regions with established pipelines, CCS hubs can become operational sooner, making them a practical near-term emissions reduction solution.

3. Model framework

This section outlines the optimization framework used for designing a cost-efficient CCS infrastructure. The framework integrates economic, environmental, and operational factors to efficiently match CO_2 emission sources with storage sinks. A Mixed-Integer Linear Programming (MILP) model optimizes source-sink matching by minimizing total CCS costs (capture, transport, storage) while ensuring compliance with storage capacity limits. Decision variables include CO_2 flow assignments, pipeline construction decisions, and hub clustering choices. The CCS-NetOptimizer code has been released as an open-source project to support further research and practical implementation. The repository contains the Python implementation, required dependencies, and a sample dataset for testing. Researchers and industry professionals can access, modify, and contribute to the project to enhance its capabilities. The source code is available at: https://github.com/aminatgy/CCS-NetOptmizer.

3.1. Overview of the optimization framework

The comprehensive workflow is illustrated in Fig. 2, detailing the step-by-step CCS optimization methodology. The figure represents data integration, where CO_2 emission points, storage sites, and pipelines are identified and incorporated into the model. Next, data processing visualizes the spatial distribution of sources and sinks, overlaying infrastructure for precise network representation. The framework then generates multiple scenarios, including full and partial redirection, alongside optimized route generation. These scenarios are refined using clustering techniques, adjusting radii to evaluate different configurations. Multiple source–sink matching scenarios were constructed by a two-factor experimental design. (i) The clustering radius r was swept from 100 to 400 km and supplied as the eps argument in DBSCAN, thereby generating seven distinct hub configurations. (ii) For each radius two independent routing logics were applied: Full redirection, in which DBSCAN members are piped through an MST prior to linkage to

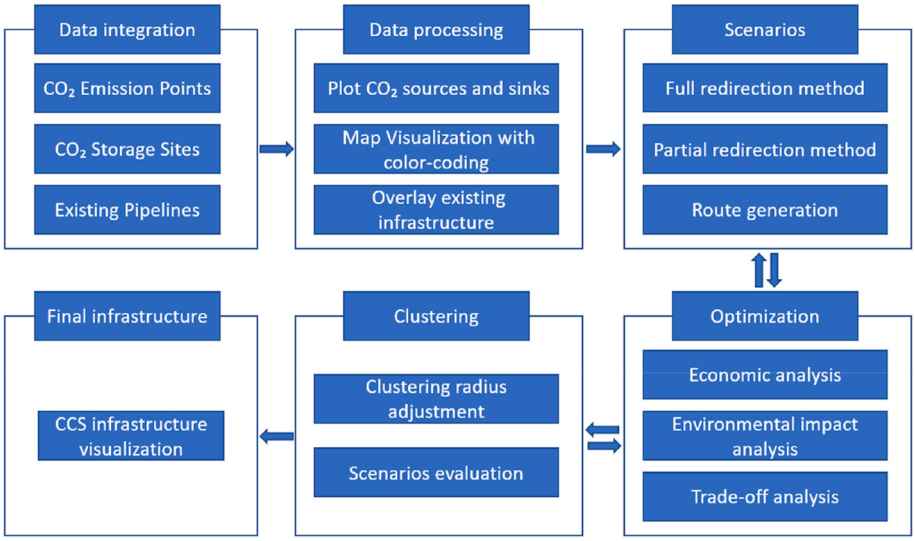


Fig. 2. CCS Optimization framework.

the single least-cost sink, and Partial redirection, in which each source (or sub-cluster) is connected to its nearest feasible sink without intra-cluster piping.

The optimization phase follows, incorporating economic analysis, environmental impact assessment, and trade-off analysis for an efficient CCS network. Finally, the optimized infrastructure is visualized, demonstrating the final CCS layout and implementation feasibility.

The combined environmental and economic objectives can now be formulated as a multi-objective optimization (MOO) problem, where EI represents the net CO_2 emissions from the CCS supply chain, and TC denotes the total cost, including capture, transportation, and storage expenses.:

$$Minimize: EI + TC \quad \text{(Eq. 1)}$$

Guided by Kazakhstan's net-zero-by-2060 target, we frame the optimization as a weighted-goal programme in which the environmental term—minimizing residual CO_2 receives a dominant weight and the cost term a secondary one (Decree of the President of the Republic, 2023). This forces the optimization to eliminate unstored emissions first and then choose the least-cost network among compliant designs. The weighted-sum approach thus provides a single, policy-aligned solution per clustering radius in one deterministic MILP run.

3.2. Economic model

There are no CCS projects in Central Asia, and globally, most are in the Oil & Gas and Chemical industries, with few in power, steel, and cement. Due to limited CCS performance and cost data across sectors, this study uses techno-economic data from US and EU studies. The technical data remains unchanged, while cost data was adapted using the CEPCI, location, and time factors for Kazakhstan (Khoyashov et al., 2024).

3.2.1. Capture cost

Capture costs ($C_{capture}$, USD) are estimated following the methodology proposed by Khoyashov et al. (2024), which incorporates capture rate factors (CRF), emission capacity (Q, $tCO2/year$), unit capture costs (C_{unit}, $USD/tCO2$), and adjustment factors for Kazakhstan.

$$C_{capture} = CRF \cdot C_{unit} \cdot Q \quad \text{(Eq.2)}$$

Table A1 in the Appendix summarizes the industrial parameters used in the optimization program (Hughes et al., 2022, 2023; James et al., 2019; Sannan et al., 2017). This study prioritizes post-combustion capture for its adaptability to existing industrial sources, with Rochelle (2009) and Rubin et al. (2012) highlighting its cost efficiency and reliability over pre-combustion and oxy-fuel methods. Using amine-based solvents like monoethanolamine (MEA), post-combustion capture is widely applied but requires significant heat energy for solvent regeneration, leading to a notable energy penalty. Rubin et al. (2012) identify this as a major barrier to large-scale deployment. While oxy-fuel combustion offers high-purity CO_2 streams, its higher capital costs make post-combustion more viable for retrofitting existing plants.

3.2.2. Transportation cost

The model assumes pipeline transport as the sole CO_2 transportation method due to its cost efficiency, safety, and scalability over alternatives like rail or truck transport, with studies estimating transport costs between 0.01 and 0.05 USD/tCO_2/km depending on pipeline diameter, length, terrain, and regional cost factors (Middleton and Bielicki, 2009; Knoope et al., 2013).

McCoy and Rubin (2008) developed an engineering-economic model for CO_2 pipeline transport, incorporating a Cobb-Douglas (Eq. (3)) function to capture the nonlinear relationship between length (L, km) and diameter (d, m) in pipeline capital costs ($C_{pipeline}$, USD). The diameter is determined using the Darcy-Weisbach equation, with the Haaland approximation applied for friction factor estimation. The pipeline capital cost model also incorporates variations in labor, materials, land, and miscellaneous costs assuming a 30-year lifespan:

$$C_{pipeline} = 1294\left(L^{0.90}d^{1.59}\right) + 30690\left(L^{0.82}d^{0.94}\right) + 8913\left(L^{1.05}d^{0.40}\right) + 24547\left(L^{0.78}d^{0.79}\right) \quad \text{(Eq. 3)}$$

Pressure booster facilities are required to maintain CO_2 flow over long distances. Their capital cost (C_{PBF}, USD) (Eq. (4)) is estimated using a formula linking cost to installed booster station power (P_{PBF}, MW):

$$C_{PBF} = 8.35P_{PBF} + 0.49 \quad \text{(Eq. 4)}$$

Booster stations are installed every 70–200 km (Solomon et al., 2024; Diabate et al., 2024; McCoy and Rubin, 2008) to maintain consistent CO_2 flow. Energy consumption across the network depends on total CO_2 flow rates and segment-specific power demands. The power (P_{PBF}, MW) required for isentropic gas compression (Eq. (5)) is determined using a thermodynamic relationship, considering the specific heat ratio of CO_2 (k), universal gas constant (R, $m3{\cdot}Pa/(mol{\cdot}K)$), average pipeline temperature (T, K), pressure ratio P_{out}/P_{in} (Pa), compressor efficiency (e), and mass flow rate (Q, kg/s), reflecting the energy needed for gas compression.

$$P_{PBF} = \frac{k}{k-1} \cdot \frac{RT}{e} \cdot \left(\left(\frac{P_{out}}{P_{in}} \right)^{(k-1)/k} - 1 \right) \cdot Q \quad \text{(Eq. 5)}$$

The total transportation cost is estimated by combining pipeline and booster costs, considering both distance and energy requirements for continuous operation.

Existing pipeline segments were modelled as reusable corridors with unrestricted tie-in, meaning CO_2 could enter and exit the repurposed network at any optimal point. This approach maximizes the potential cost savings from pipeline reuse, consistent with prior graph-based optimization studies. Under this assumption, existing infrastructure enables up to 44 % reduction in unit transport cost compared to full greenfield build-out (Nnamdi and Moghanloo, 2023). In our model, the full reuse potential is applied wherever existing pipelines align with optimal source–sink paths.

3.2.3. Storage cost

Recent studies estimate CO_2 storage costs in onshore saline aquifers between $4–$45 per tonne, influenced by transport distance, operation scale, monitoring, geology, and cost variability (Smith et al., 2021). The Zero Emissions Platform (2019) reports €1–€7 per tonne for onshore depleted fields and €6–€20 for offshore aquifers, emphasizing reservoir characteristics and injectivity. McCoy and Rubin (2008) developed an engineering-economic model to improve cost estimation by accounting for regional and project-specific variations. Our study applies McCoy's model using a single-region approach for geological analysis. The storage cost model includes drilling and completion (DC), injection equipment (IE), and compressor costs.

DC and IE costs (C_{DC+IE}, *USD*) follow the Lewin & Associates model, estimating well drilling and completion via a regression formula adjusted for regional variations. Injection well costs are assumed comparable to water injection wells (McCoy and Rubin, 2008; Bock et al., 2003) and adjusted for economies of scale using EIA regional cost factors. The scaling equation (Eq. (6)) follows a power-law approach, reducing costs as more wells are drilled, with maximum efficiency at 21 wells per site. The coefficient a_1 represents the baseline cost factor, while a_2 determines the exponential cost increase with depth (*D, m*), reflecting higher expenses for deeper wells. These coefficients vary by region, adjusted for geographic similarity in this study.

$$C_{DC+IE} = a_1 \cdot exp(a_2 D) \quad \text{(Eq. 6)}$$

The capital cost of the compressor system (Eq. (7)) follows the International Energy Agency (IEA) model, which estimates costs based on installed booster station power, the same as in the case of transportation.

$$C_{injection} = 8.35 P_{inj} + 0.49 \quad \text{(Eq. 7)}$$

The model accounts for the power (Eq. (8)) required to inject CO_2 at the designed flow rate (Q, m^3/s), static (*Ps*,*Pa*), and friction pressure (*Pf*, *Pa*), with adjustments made for aquifer depth and wellhead pressure constraints of the pump efficiency (e) assumed to be 85 %. Where ρ, (kg/m^3) is CO_2 density, g, (9.81 m/s) is gravitational acceleration, f is the friction factor, D, m is the aquifer depth, $\nu(m/s)$ is the injection velocity, and d_w *(m)* is the well diameter.

$$P_{inj} = Q \frac{\Delta Ps + \Delta Pf}{e} = Q \frac{\left(\rho g D + \frac{f D \rho}{2 d_w} \nu^2 \right)}{e} \quad \text{(Eq. 8)}$$

3.3. Environmental model

CCS technology reduces greenhouse gas emissions and contributes to emissions through its supply chain. Using the emission factor method, total supply chain emissions scale linearly with the abatement target, while the relative shares of capture, compression, transport, injection, and monitoring remain unchanged (Nie et al., 2023). The environmental objective is to minimize the total net emission abatement, which can be formulated as:

$$EI_{total} = E^{capture} + E^{transport} + E^{storage} \quad \text{(Eq. 9)}$$

where EI_{total} represents the total net emission abatement, and $E^{capture}$, $E^{transport}$, and $E^{storage}$ (tCO_2/year) corresponds to emissions from CO_2 capture, transport, and storage, respectively.

The net CO_2 abated can be expressed as:

$$CO_2{}^{abated} = CO_2{}^{source} - EI_{total} \quad \text{(Eq. 10)}$$

where $CO_2{}^{abated}$ (tCO_2/year) is the amount of CO_2 effectively removed from the atmosphere, $CO_2{}^{source}$ (tCO_2/year) refers specifically to emissions from the source plant before any CCS operations, and EI_{total} accounts for the cumulative emissions from capture, transport, and storage phases. This equation reflects the trade-off between the amount of CO_2 captured and the emissions associated with the CCS infrastructure, highlighting the need to optimize operational parameters to maximize net reductions.

The emissions associated with the capture phase are determined by Eq. (11), where *CRF* is the capture rate factor obtained from Table A1. This equation accounts for the fraction of CO_2 successfully captured and removed, with the remainder contributing to total emissions.

$$E^{capture} = (1 - CRF) \cdot CO_2{}^{source} \quad \text{(Eq. 11)}$$

The transportation phase contributes to emissions primarily through fugitive CO_2 emissions along pipeline routes and the electricity consumption of pressure boosters. Fugitive emissions are treated as a constant rate, defined in terms of emissions per kilometer per tonne of CO_2 transported (Eq. (12)), where the $EF_{f-pipeline}$ (tCO_2/km/year) is the factor of pipeline fugitive emission from IPCC et al., 2023 estimated at 1.4 tCO_2/km/year and L is the pipeline length, m.

$$EI^{transport}_{pipeline-fugitive} = EF_{f-pipeline} \cdot L \quad \text{(Eq. 12)}$$

Electricity-related emissions from compressor stations are estimated using the emission factor of Kazakhstan's electricity grid, calculated as:

$$EF_e = \sum_{i=1} nEF \cdot ESS \quad \text{(Eq. 13)}$$

Where EF_e ($kgCO_2e$/kWh) is the weighted average electricity emission factor, *EF* ($kgCO_2e$/kWh) is the emission factor of the electricity source, and ESS is the share of the electricity mix from the source. Based on Kazakhstan's electricity grid mix, the weighted average electricity emission factor is calculated to be 0.882 $kgCO_2e$/kWh.

For pressure boosters, the carbon footprint was calculated based on the electricity required to maintain CO_2 flow along the pipeline. The energy consumption of pressure boosters and injection compressors is converted into emissions using:

$$EI^{transport}_{pbf-energy} = EF_e \cdot P_{pbf} \quad \text{(Eq. 14)}$$

$$EI^{storage}_{inject-energy} = EF_e \cdot P_{inj} \quad \text{(Eq. 15)}$$

where P_{pbf} and P_{inj} (kWh/year) are the power requirements for pressure boosters and injection compressors, respectively.

By integrating these factors, the overall environmental impact assessment captures both direct and indirect emissions associated with the transport and storage phases, providing a comprehensive view of the carbon footprint of CCS infrastructure.

3.4. Model inputs

3.4.1. Emission capacity in Kazakhstan

Kazakhstan has significant natural gas reserves, with production growing 4 % annually from 2010 to 2020, reaching 55.1 billion m^3 in gross and 34.8 billion m^3 in commercial output by 2020. The country

relies heavily on fossil fuels, with 82 % of its 23.6 GW power capacity sourced from them, making the power sector responsible for 80 % of total CO_2 emissions.

Effective CCS deployment requires understanding emission sources. Major CO_2 emitters include 39 pulverized coal power plants (87.91 Mt CO_2 annually), 19 natural gas power plants (11.59 Mt), and 6 gas processing plants (12.01 Mt). Industrial contributors include 11 cement plants (8.06 Mt), 9 iron and steel plants (24.91 Mt), and 4 oil refineries (4.05 Mt), totaling 148.53 Mt CO_2 annually.

Key emission regions include Pavlodar, Ekibastuz, and Karaganda (coal-fired power and steel plants); Atyrau, Aktobe, and Mangystau (oil refineries and gas processing); Shymkent and Taraz (cement production); and Central and Western Kazakhstan (six major iron and steel plants near ore deposits). These regions require a strategic CCS hub design to optimize transport logistics and reduce infrastructure costs.

3.4.2. Storage capacity in Kazakhstan

Kazakhstan's CO_2 storage potential has been assessed using the USDOE method, aligned with the Carbon Sequestration Leadership Forum (CSLF) framework (Goodman et al., 2011; Bachu, 2008). Effective CO_2 storage capacity is estimated using:

$$M_{CO2} = E \cdot A \cdot h \cdot \varphi \cdot \rho \quad \text{(Eq. 16)}$$

Where E is the storage efficiency factor, A is the total area, h is the formation thickness, ϕ is the porosity, and ρ is the CO_2 density under reservoir conditions. Abuov et al. (2020) identified six major basins—Precaspian, Mangyshlak, South-Torgay, Ustyurt, Chu-Sarysu, and Zaysan—as the most suitable for CCS, ranking them by containment potential and feasibility.

The total CO_2 storage capacity of Kazakhstan is estimated at approximately 583 Gt, with 539 Gt located in the Precaspian, Mangyshlak, South-Torgay, and Ustyurt basins, where most injected CO_2 is expected to be stored in hydrodynamic traps (Abuov et al., 2020). The Precaspian Basin is the most promising, providing the highest containment potential and large-scale injection feasibility. This basin features deep carbonate reservoirs and extensive salt domes that serve as natural seals, ensuring minimal leakage risks (Abuov et al., 2020).

Seventeen optimal storage sites with a combined capacity of 5784 Mt CO_2 have been identified (Table A2). T17 and T15 (Chu-Sarysu Basin) offer high storage capacities, T16 (South-Torgay Basin) has thick formations and favorable porosity, and T14 and T13 (Zaysan Basin) provide well-defined reservoir properties. All other identified sites belong to the Precaspian Basin, which has extensive storage potential and favorable geological conditions.

3.5. Model constraints

The mathematical formulation of the CO_2 source and sink matching tool was introduced to optimally assign CO_2 sources to sinks, considering multiple constraints, including distance minimization, sink capacity limits, and the possibility of multiple routes between a source and sink.

Each CO_2 source is assigned to a single sink to simplify allocation and enhance pipeline utilization efficiency (Eq. (17)), where $x_{ij} = 1$ if source i is connected to sink j.

$$\sum_{i=1}^{m} x_{ij} = 1, \forall i \in s \quad \text{(Eq. 17)}$$

Storage limits prevent overloading of sinks (Eq. (18)), where C_i represents captured CO_2 at source i, and D_j represents the maximum storage capacity of sink j.

$$\sum_{i=1}^{n} C_i \cdot x_{ij} \le D_j, \forall j \in T \quad \text{(Eq. 18)}$$

Optimal pipeline routes are selected to minimize total transportation distance and costs (Eq. (19)), where y_{ijk} determines whether route k is used between source i and sink j. Infrastructure clustering is implemented using Density-Based Spatial Clustering (DBSCAN), which enhances efficiency by grouping emission sources into shared transport networks. This method minimizes transport distances and costs while optimizing storage utilization.

$$\sum_{k \in routes(i,j)} d_{ijk} y_{ijk} = \min\left(d\left(s_i, t_j\right)\right) \forall i \in s \quad \text{(Eq. 19)}$$

Pipeline distances were approximated as great-circle paths and therefore do not yet account for (i) elevation changes that raise recompression energy, (ii) land-use constraints such as protected areas and urban zones, or (iii) routing optimization via least-cost path algorithms. International cost reviews (Bui et al., 2018) indicate that terrain complexity and water crossings can inflate capital expenditure by 10–30 %. Future work will couple the optimization engine to a high-resolution GIS layer so that terrain penalties, land-acquisition costs, and routing constraints are explicitly reflected in both the economic and environmental objectives.

3.6. Uncertainty analysis

To propagate parametric uncertainty into the transportation-cost model, we implemented a Monte-Carlo (MC) simulation that mirrors the pipeline-cost framework of McCoy and Rubin (2008). For other costs, the uncertainty range of inputs for the Kazakhstan case is not available. Five inputs with the largest influence on levelized cost were treated as stochastic: (i) the capital-recovery factor (CRF, U[0.10, 0.20] yr^{-1}), (ii) the annual O&M rate (U[2,150, 4350] US $ km^{-1} yr^{-1}), (iii) the effective pipeline-capacity factor (U[0.50, 1.00]), (iv) a capital-cost escalation multiplier (U[0.75, 1.25]) that captures engineering-estimate uncertainty, and (v) the compressor inlet pressure (U[12, 15] MPa). For each of 5000 trials, the script randomly drew a value from each distribution, re-executed the original routing routines for both full (F) and partial (P) redirection, and computed the annualized transport cost per tonne of CO_2. The empirical distributions were summarized by the median (P50) and the 5th–95th percentiles, which form the 90 % confidence intervals visualized as error bars. This MC block was coded in Python 3.12 using NumPy's uniform sampler, providing a statistically robust envelope of likely transportation costs without altering any deterministic calculation steps.

4. Optimization results

4.1. Comparison analysis of the scenarios

Clustering sources within a 100 km (Fig. 3a) radius minimizes total system costs by up to 25 % compared to dispersed storage strategies. This cost reduction is due to pipeline consolidation and shared capture facilities, lowering per-tonne transportation and injection costs. However, beyond 240 km (Fig. A2a), transportation costs rise disproportionately due to pipeline pressure losses and additional compressor stations.

Full redirection is more cost-effective in Kazakhstan than partial redirection since most CO_2 storage sinks have large capacities but are far from emission sources, except in the western region, where sources and sinks are closely aligned (Fig. A1, Appendix). In full redirection (Fig. A1a), all emissions from a full storage site are redirected to the nearest available sink, optimizing pipeline use and improving storage efficiency. In partial redirection (Fig. A1b), emissions are distributed across multiple sinks, allowing some sources to remain at their initial sites, increasing flexibility but potentially raising transport costs. Developing CCS hubs with multiple emitters enhances cost efficiency in Kazakhstan (Khoyashov et al., 2024). Fig. 3a shows that a 100 km clustering scenario creates compact hubs, reducing pipeline length and

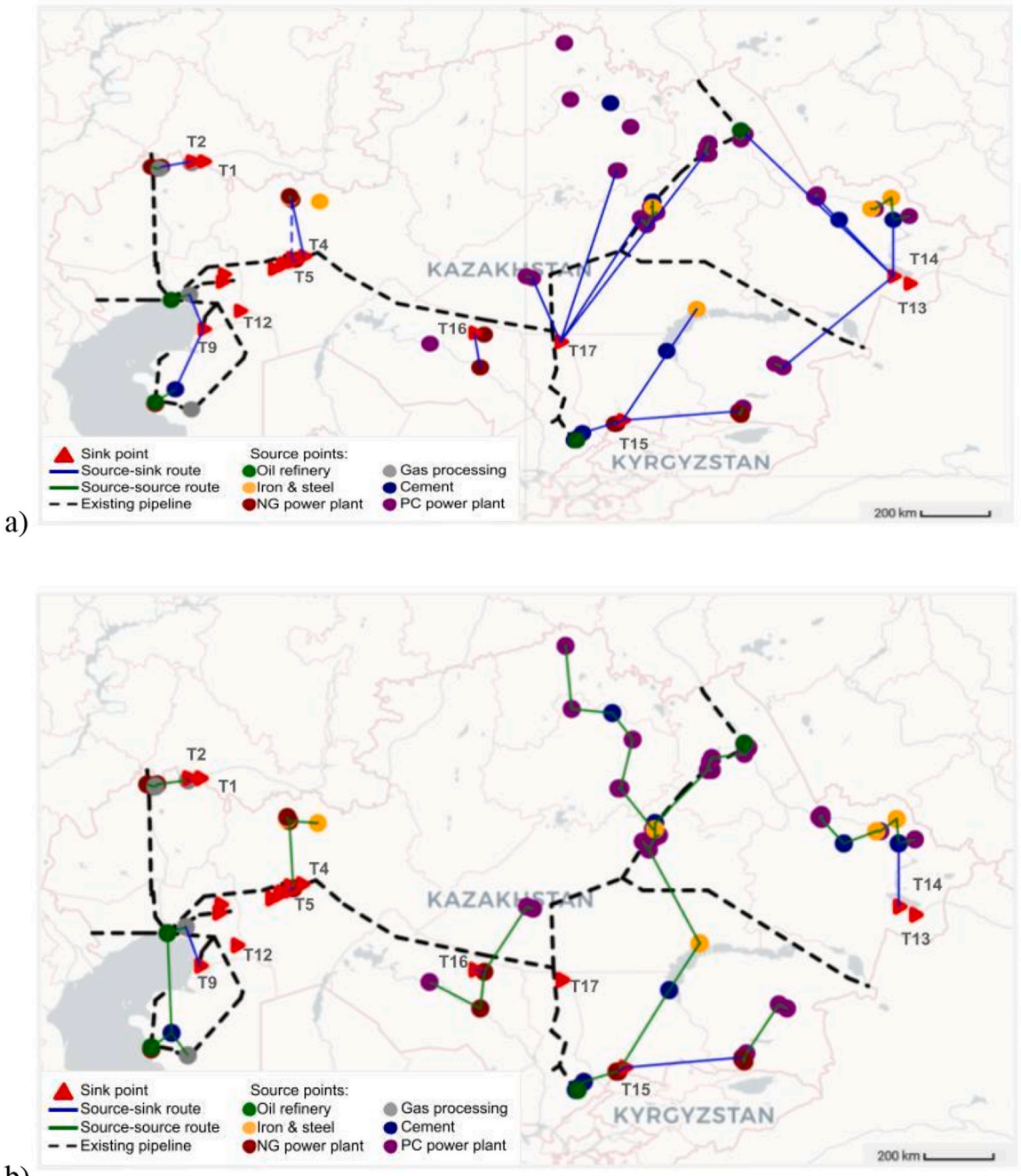


Fig. 3. CCS hub system with a) the lowest total cost (100 km, full redirection) and b) the lowest net emission abatement (360 km, full redirection).

transport costs, whereas Fig. 3b illustrates a 360 km clustering scenario with dispersed storage sites, leading to longer transport distances but lower per-tonne emissions.

Fig. 4 shows the total cost and CO_2 abated across different scenarios. Full redirection achieves the lowest cost per tonne of CO_2 stored by optimizing pipeline routing and reducing compression energy demands. The 100 km clustering scenario has the lowest total cost, with full redirection at 201.38 USD/tCO_2 and partial redirection at 207.61 USD/tCO_2. However, transportation costs at 100 km are higher, reaching 91.85 USD/tCO_2 (full redirection) and 96.48 USD/tCO_2 (partial redirection). As the clustering radius increases, transportation costs rise due to longer pipeline distances and higher compression requirements, reaching 197.23 USD/tCO_2 at 400 km. While total unit costs remain between 201 and 306 USD/tCO_2, they are higher than some global CCS hubs, such as Norway's Northern Lights ($80–150 per tonne) (Zero Emission Platform, 2021). In contrast, U.S. Gulf Coast studies estimate CCS costs at $60–120 per tonne, highlighting the need for further cost optimization in Kazakhstan.

One of the main reasons for the higher costs in Kazakhstan's CCS hub is the long transport distances between emission sources and available storage sites. Unlike Norway and the U.S., which have dense industrial clusters close to storage locations, Kazakhstan's geographically dispersed industrial sources necessitate extensive pipeline infrastructure, significantly increasing transportation expenses. Additionally, Kazakhstan's CCS hub includes multiple industry sources (e.g., power plants, cement factories, chemical plants), which require different CO_2 separation processes, leading to higher capture costs than hubs primarily handling high-purity CO_2 from natural gas processing.

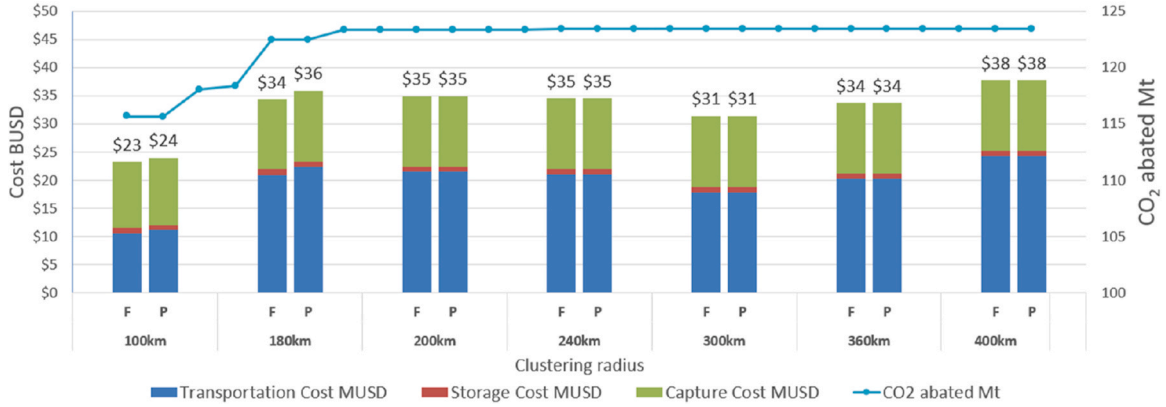


Fig. 4. CO_2 abated and total cost per clustering radius for full (F) and partial (P) redirections of the CCS Hub system.

Further comparison with the work of Khoyashov et al. reveals that the Atyrau hub was found to have a low total cost of CO_2 capture due to the dominance of natural gas processing plants, which offer high-purity CO_2 streams, reducing separation costs. Conversely, the North hub, containing numerous coal-fired power plants, faced higher capture costs due to low CO_2 purity and greater transportation distances, necessitating multiple pressure boost facilities. The optimization program of 360 km clustering radius (Fig. 3b) resulted in similar hub configurations for Almaty, Oral, Atyrau, and Aktau, as the sources and sinks in these regions are closely clustered. However, other hubs, while centered similarly, differ in source-sink connections due to updated sink capacity estimations, leading to variations in clustering radii and route optimizations.

4.2. Cost component analysis

The cost breakdown highlights capture costs as the dominant expense, accounting for 50.47 % of total costs at a 100 km radius, but decreasing to 36.17 % at 180 km due to the increasing contribution of transportation costs (Fig. 4). This shift suggests that as transport distances increase, pipeline infrastructure and compression costs can be the primary cost drivers rather than capture expenses. Transport costs do not increase linearly but stabilize, suggesting economies of scale in pipeline routing efficiency.

Transportation costs exhibit an increasing trend, representing 45.61 % at 100 km and 61.01 % at 180 km, emphasizing the infrastructure challenges of expanding clustering radii. Compared to existing CCS projects globally, Kazakhstan's transport costs are on the higher side, reflecting the long distances between emission sources and available storage sites. The International Energy Agency (IEA, 2021) suggests that transportation costs typically range from $10–40 per tonne, whereas Kazakhstan's values exceed $90 per tonne at short radii and reach nearly $200 per tonne at 400 km. This higher transportation cost can be attributed to the need for multiple booster stations and high compression requirements over long distances. Unlike CCS hubs in the U.S. Gulf Coast, which benefit from short, high-volume pipelines, Kazakhstan's CCS network requires long-distance CO_2 transport with frequent pressure boosts, increasing operational costs.

The integration of existing pipeline infrastructure significantly reduces transportation costs by leveraging pre-established routes. Fig. 5 illustrates that considering existing pipelines lowers transport costs across all clustering radii, with the most substantial savings occurring in mid-range clustering distances. For instance, at 100 km clustering, transportation costs drop from 11 MUSD to 10 MUSD, while at 400 km clustering, the reduction is from 24 MUSD to 23 MUSD. This cost mitigation effect underscores the importance of reusing existing pipeline infrastructure wherever feasible to optimize CCS deployment costs.

Net CO_2 abatement rises to about 123 Mt by a 200 km clustering radius and then plateaus because most large emitters are already captured, and the extra energy for longer pipelines offsets the small additional volumes (Fig. 6). Extending the network beyond 200 km, therefore, raises costs without meaningful environmental benefit, making ~200 km the pragmatic limit.

Narrow bands at small radii. At 100 km, the ±90 % band is only ±20 % of the median value ($11 B vs. a spread of $9–13 B). This reflects the short pipe lengths and correspondingly low sensitivity to pressure-loss assumptions. Broadening uncertainty toward 400 km. The band widens to ±40 % around the median at 400 km. Longer pipeline length amplifies the joint uncertainty of friction losses (hence inlet-pressure requirements) and the number of booster stations, explaining the taller whiskers. Across all radii, the median bars for full and partial redirection differ by < 6 %, and their confidence intervals largely overlap. Thus, the choice between Full and Partial is driven more by strategic factors (e.g., storage logistics) than by expected transport-cost

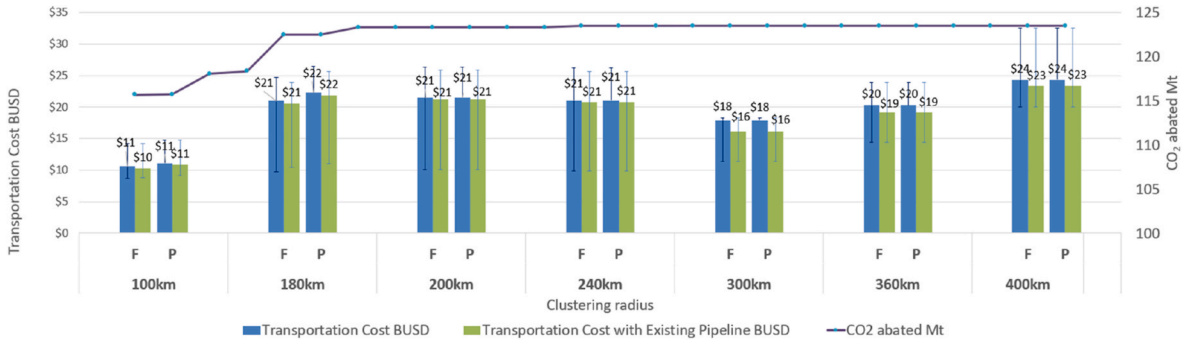


Fig. 5. CO_2 abated and total transportation cost per clustering radius for full (F) and partial (P) redirections of the CCS Hub system with the existing pipeline scenario.

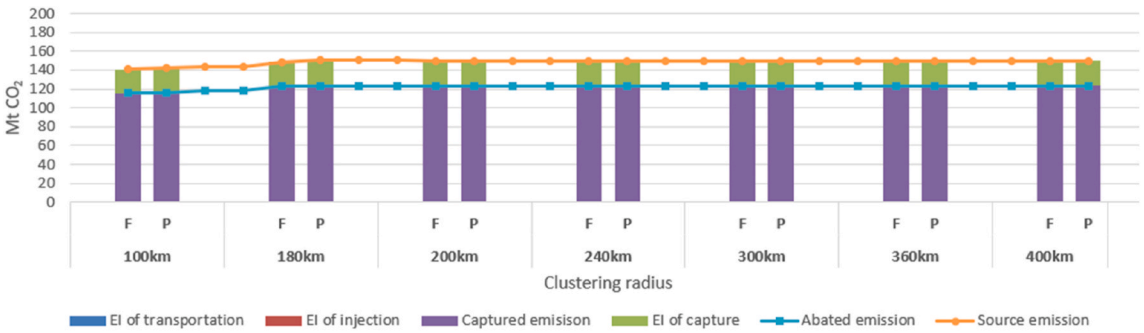


Fig. 6. Environmental impact per clustering radius for full (F) and partial (P) redirections of the CCS Hub system.

savings.

Overall, the total distance of the intersections with the existing pipelines ranges between 285 and 2800 km (Fig. 7). The longest intersection is observed starting at a 300 km clustering radius (Fig. A2b) because the greatest central North Hub's transportation route lies through the existing route and the optimized program identified the North hub as requiring multiplication into five parallel pipeline systems starting from a clustering radius of 270 km. This adjustment was necessary as pipeline transportation capacity was exceeded, and no alternative sinks, except those in the central region, were available to redirect emissions. In contrast, previous studies designed only four parallel pipelines, which excluded several northern industrial emission zones. This integration of automated optimization tools has proven essential in refining logistical planning and ensuring the cost-effective deployment of CCS infrastructure.

4.3. Environmental impact analysis

The environmental footprint of CCS hubs is dominated by CO_2 capture emissions, which contribute 99.95 % of total system emissions across all clustering radii (Fig. 6). Transport-related emissions remain below 0.06 %, while injection-related emissions are negligible (<0.0003 %). Properly managed CO_2 storage sites have leakage rates below 0.01 % per year, ensuring long-term containment. Potential risks include wellbore integrity failures and caprock permeability, which can be mitigated through continuous pressure monitoring and injection control. However, the selection of storage sites significantly influences environmental performance. The study shows that T2, T5, T9, T14, T15, T16, and T17 were the primary sinks used, with T15 and T17 (Chu-Sarysu Basin) offering high storage capacities, while T16 (South-Torgay Basin) provides thick formations with favorable porosity (Table A2). The depth and porosity of selected storage sites impact injection-related energy consumption. T16 has a 1200 m thickness, reducing the required injection pressure and potentially lowering the energy footprint compared to smaller reservoirs. Conversely, deeper formations with low porosity and density require higher injection pressures, increasing energy consumption and emissions.

Transportation emissions show a slight variation with clustering radius, increasing marginally at higher radii due to extended pipeline lengths and additional booster station requirements. Long transport distances to high-capacity sinks such as T17 and T15 increase CO_2 compression requirements, leading to higher energy demands for maintaining pipeline pressure. In contrast, closer storage sites with sufficient capacity may reduce overall transportation emissions, improving CCS efficiency. This trade-off highlights the importance of balancing economic feasibility with environmental performance when selecting sink locations. These findings emphasize that while CCS plays a crucial role in mitigating CO_2 emissions, its overall environmental footprint is strongly linked to capture efficiency and energy consumption. Optimizing capture technologies, integrating low-emission energy sources, and improving pipeline efficiency will be key to enhancing the sustainability of large-scale CCS deployment.

4.4. Discussion on how Kazakhstan can achieve this planned infrastructure

The optimized CCS hub design for Kazakhstan demonstrates the feasibility of efficient CO_2 source-sink matching and transportation. However, large-scale implementation requires substantial government support to drive industry participation and secure long-term investment. Policymakers must establish clear regulatory and financial frameworks to mitigate the high upfront costs and risks associated with early-stage projects. Government interventions, such as subsidies, risk-sharing mechanisms, carbon pricing tools, carbon credits, and tax incentives, are crucial for attracting private sector investment. Linking CCS incentives with enhanced oil recovery, as seen in other regions, could provide immediate economic benefits while advancing decarbonization efforts (IEA, 2020).

Currently, Kazakhstan lacks a dedicated CCS legal framework, making regulatory reform essential for large-scale deployment. Amending the Subsoil Code to explicitly permit CO_2 storage, revising the Environmental Code to classify CCS as a climate mitigation activity, and

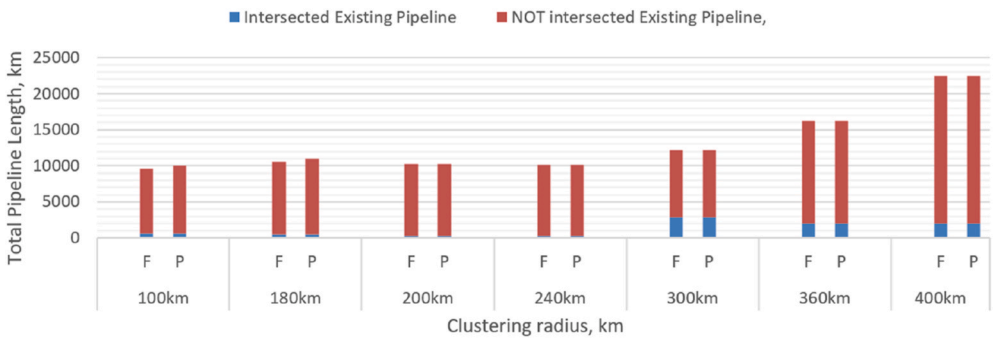


Fig. 7. Total pipeline length per clustering radius for full (F) and partial (P) redirections of the CCS Hub system.

developing standalone CCS legislation would enhance legal certainty for investors. Drawing on global best practices, such as the European Union's CCS Directive, can guide the creation of structured permitting systems, long-term liability frameworks, and clear monitoring requirements for safe and efficient CO_2 storage.

Kazakhstan can also benefit from international collaboration. Several emerging economies—including China, India, Mexico, Saudi Arabia, and South Africa—are in the process of crafting CCS regulations (Jiang et al., 2020). Engaging with these peer initiatives and aligning domestic standards with ISO/TC 265 will grant access to technical know-how, multilateral funding, and best-practice monitoring protocols (ISO, 2024).

A structured carbon credit system could increase financial incentives for CCS, assigning higher credit values to CO_2 capture and storage, making projects more attractive and globally competitive (IEA, 2021). To support effective implementation, government-backed pilot projects can demonstrate feasibility, validate economic and environmental benefits, and enhance local expertise through training programs.

Transparent communication with industry stakeholders and the public is essential for broad acceptance and establishing CCS as a key decarbonization strategy. With vast geological storage potential and a strong oil and gas industry, Kazakhstan is well-positioned to become a regional CCS leader. However, achieving this requires coordinated efforts from the government, industry, and international partners. By fostering investment, streamlining regulations, and implementing targeted financial mechanisms, Kazakhstan can transition from a technical CCS model to a real-world solution, accelerating the shift toward a low-carbon economy.

5. Conclusions

The optimization of Kazakhstan's CCS infrastructure through source-sink matching and clustering strategies presents significant opportunities for reducing costs and enhancing environmental sustainability. This study highlights the impact of clustering radii on CCS system efficiency, demonstrating that shorter radii (100 km) provide the lowest-cost solutions due to efficient infrastructure utilization and minimal pipeline requirements, while longer radii (360 km) prioritize environmental benefits at the expense of higher transport costs. Key findings indicate that integrating existing pipeline infrastructures significantly enhances cost efficiency, particularly in shorter clustering radii, whereas longer radii require additional investments in transport infrastructure. The results also emphasize the necessity of balancing cost and environmental objectives, with intermediate clustering radii (240 km) offering a trade-off between affordability and emissions reductions. While regional variations influence cost stabilization points, the overall trends align with international benchmarks on CCS hub formation and transport cost efficiency. CCS hub optimization involves complex trade-offs between cost and environmental efficiency. The 100 km radius is ideal for projects with financial constraints, offering the lowest cost per tonne of CO_2 abated. Meanwhile, 240 km–300 km scenarios provide a balanced approach, stabilizing costs while minimizing emissions. Future research should focus on refining CCS infrastructure optimization by incorporating dynamic factors such as regulatory frameworks, evolving CO_2 pricing mechanisms, and advancements in capture technologies. Additionally, expanding the scope to include real-time adaptive clustering methods and dynamic storage capacity adjustments would further enhance the robustness and applicability of CCS planning models to diverse local and global CCS hub cases.

CRediT authorship contribution statement

Amina Togay: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Gaini Serik:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Woojin Lee:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to improve the readability of some sentences. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2025.146563.

Data availability

Data will be made available on request.

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